M13-C3 — H7a `asm_intro(relto)` vs `asm_intro(sechk)` under brave nd

H7a run / analysed / not a Bucket 3 claim

1 Status

H7a run / analysed. Bounded implementation-behaviour investigation of inherited ABALearn `asm_intro` under brave nd on H0 and H3 `n30_seed42` (7 cells). Markdown twin: `h7a_relto_vs_sechk.md`.

Lead: on all seven pairs both arms `solved`, but every delta differs — a genuine `relto`↔`sechk` control-flow fork that systematically changes what is learned. Under sechk, deltas latch onto the first BK predictor and close with an α-chain on that variable alone; under relto, later BK covariates more often enter the contrary layer. Config `nd_brave_sechk` is experimental (not `ecai2024_sechk`; not a published ECAI method). H0–H6 / H4b closed. **Not a Bucket 3 claim.** H7b analysed separately (`h7b_cautious_relto_vs_sechk.md`).

2 Research question

Holding fixture, frozen sample, encoding, brave semantics, nd folding, and all other learner options constant: how does changing `asm_intro(relto)` to `asm_intro(sechk)` affect what ABALearn learns and how search proceeds under brave nd?

3 Main finding — the relto ↔ sechk contrast

3.1 They diverge

On all seven pairs (`ecai2024/relto` vs `nd_brave_sechk/sechk`): both arms `solved`; all deltas differ; BK/examples/data byte-identical within each pair.

Especially for mechanism-relevant c:

Cell	Relto	Sechk
H0 c	gate a + contrary `b_val_0`	gate a + α-nest on a only (no b)
H3 c	gate a + contraries b/d (+ nest)	gate a + α-nest on a only (no b/d)

For child c, sechk is less covariate-informative than relto here.

3.2 Why they diverge

After failed post-fold entailment in `gen.pl`:

- **Relto:** `exists_assumption_relto`; after `KO`, search can backtrack to another fold literal — how later covariates enter.
- **Sechk:** always mint provisional α, then `gen4`→`gen5`/`gen6`; often never reaches those alternative folds.

First fork (every pair):

```
relto:  looking for assumption relative to
sechk:  generating NEW assumption: alpha_1(A)
```

Genuine control-flow, not cosmetic logging.

3.3 Consistent pattern (7/7 sechk)

Every sechk delta: (i) exactly one observed BK predictor — the first in that target's BK order; (ii) α-chain on that variable's `*_val_*` literals; (iii) no later BK covariates in the contrary layer. Phrase: first-BK latch + α-chain under sechk vs richer contrary covariates under relto (not "always gate on a": sechk target a uses only b).

Cross-fixture: H0 sechk a byte-identical to H3 sechk a; H0 sechk b to H3 sechk b — shared template, *not* shared causal role of a.

4 Setup

Relto: `configs/ecai2024_config.pl`. Sechk: `configs/nd_brave_sechk_config.pl` (experimental). Intervention: only `asm_intro`. Held fixed: brave, nd, steps 10, any, all, `check_ic`. Fixtures: H0 `m13_bucket3_binary_collider_and`; H3 `m13_bucket3_binary_bd_and_lead_a`; sample `n30_seed42`.

Role discipline: only c is a desirable det. mechanism target; H0 a is a parent of c but exogenous as learning target; H3 a is *isolated*; role-peer of H0 a is H3 b. `solved` $\neq$ mechanism recovery.

Paths: `.../<fixture>/{ecai2024,nd_brave_sechk}/n30_seed42/`.

5 Seven-cell paired summary

Cell	BK order	out	Δ=?	sechk sole	relto also
H0 a	b, c	both sol.	no	b	b+c
H0 b	a, c	both sol.	no	a	a+c
H0 c	a, b	both sol.	no	a	a+$\mathbf{b}$
H3 a	b, c, d	both sol.	no	b	b+c/d
H3 b	a, c, d	both sol.	no	a	a+c/d
H3 c	a, b, d	both sol.	no	a	a+$\mathbf{b}/\mathbf{d}$
H3 d	a, b, c	both sol.	no	a	a+b/c

6 Primary contrast excerpts

H0 c relto:

```
c(A) :- alpha_1(A), a_val_1(A).
c_alpha_1(A) :- b_val_0(A).
```

H0 c sechk:

```
c(A) :- alpha_1(A), a_val_1(A).
c_alpha_1(A) :- alpha_2(A), a_val_1(A).
c_alpha_2(A) :- alpha_1(A), a_val_1(A).
```

H3 c: relto retains b/d in contraries; sechk α-nest on a only. H3 a (isolated) vs H3 b (parent): sechk sole vars b vs a; H0/H3 sechk a (resp. b) deltas byte-identical — not shared causal role of a.

7 What changed / did not / left open

Changed: repair path; every delta; contrary covariates (esp. loss of b/d on c under sechk). Unchanged: brave+nd bundle otherwise; inputs within pairs; `solved` outcomes here. H7b cautious nd is run / analysed separately (`h7b_cautious_relto_vs_sechk.md`); this H7a record remains brave-only.

8 Boundaries / non-claims

- Not a Bucket 3 claim.
- `solved` $\neq$ causal / mechanism recovery.
- Not a ranking of sechk vs relto as generally better.
- Pattern on these tables $\neq$ universal law.
- Do not conflate H3 a (isolated) with H0 a (parent).
- Do not call `nd_brave_sechk` an ECAI-paper config.
- H7b analysed separately; H0–H6 / H4b readings preserved.

9 Next

H7a documentation complete. H7b analysed (`h7b_cautious_relto_vs_sechk.md`). Still **no Bucket 3 claim**.